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Homework 7

1.

Example 7.4. As an extension of Example 7.1, consider the multivariate normal density

$$(X_1, X_2, \ldots, X_p) \sim \mathcal{N}_p\left(0, (1-\rho)I + \rho J\right), \tag{7.5}$$

where I is the $p \times p$ identity matrix and J is a $p \times p$ matrix of ones. This is a model for *equicorrelation*, as $\mathrm{corr}(X_i, X_j) = \rho$ for every i and j. Using standard formulas for the conditional distributions of a multivariate normal random variable (see, for example, Johnson and Wichern, 1988), it is straightforward but tedious to verify that

$$X_i | x_{(-i)} \sim \mathcal{N}\left(\frac{(p-1)\rho}{1+(p-2)\rho}\bar{x}_{(-i)}, \frac{1+(p-2)\rho-(p-1)\rho^2}{1+(p-2)\rho}\right),$$

where $x_{(-i)} = (x_1, x_2, \ldots, x_{i-1}, x_{i+1}, \ldots, x_p)$ and $\bar{x}_{(-i)}$ is the mean of this vector. The Gibbs sampler that generates from these univariate normals can then be easily derived, although it is useless for this problem (Exercise 7.5). It is,

however, a short step to consider the setup where the components of the normal vector are restricted to a subset of $\mathbb{R}^p$. If this subset is a hypercube,

$$\mathfrak{H} = \prod_{i=1}(a_i, b_i),$$

then the corresponding conditionals simply are the normals above restricted to (a_i, b_i) for $i = 1, \ldots, p$ (in which case an exact algorithm such as `sadmvn` can be used). For more complex constraints, a Gibbs sampler is however (almost) required, as exact solutions do not exist. This Gibbs sampler is still based on normal full conditionals, which are now restricted to subsets of the real line and thus easily simulated (Exercise 2.22). ◀

Exercise 7.5 Given the normal target $\mathcal{N}_p(0, (1-\rho)I + \rho J)$:

a. Write a Gibbs sampler using the conditional distributions provided in Example 7.4. Run your R code for $p = 5$ and $\rho = .25$, and verify graphically that the marginals are all $\mathcal{N}(0, 1)$.
b. Compare your algorithm using $T = 500$ iterations with `rmnorm` described in Section 2.2.1 in terms of execution time.
c. Propose a constrained subset that is not a hypercube, and derive the corresponding Gibbs sampler. (*Hint*: Consider, for example, a constraint such as $\sum_{i=1}^{m} x_i^2 \leq \sum_{i=m+1}^{p} x_i^2$ for $m \leq p-1$.)

(a)

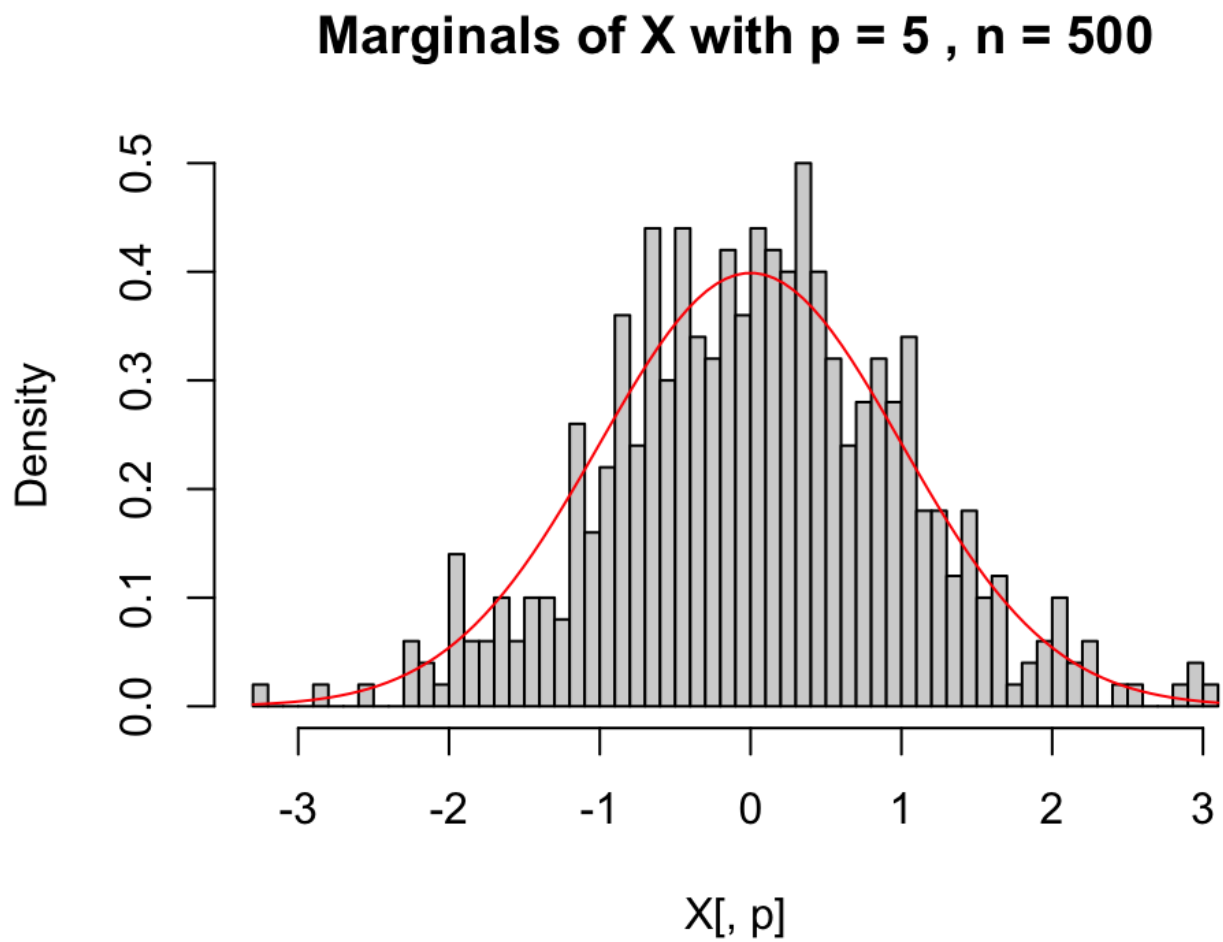
We are given the target distribution $X \sim N_p(0, (1-\rho)I + \rho J)$. From Example 7.4, the conditional distribution of X_i given all other components is said to be

$$X_i | x_{(-i)} \sim N\left(\frac{(p-1)\rho}{1+(p-2)\rho}\bar{x}_{(-i)}, \frac{1+(p-2)\rho-(p-1)\rho^2}{1+(p-2)\rho}\right)$$

which we'll use in the code below, following the algorithm of drawing conditionally based on previously drawn values.

```
> # 1. (a)
> n = 500
> p = 5
> rho = 0.25
> gibbs_sample = function(n, p, rho) {
+   # Precomputed constants for conditional distribution from Example 7.4
+   conditional_mean_coeff = ((p - 1) * rho) / (1 + ((p - 2) * rho))
+   sigma_squared = (1 + ((p - 2) * rho) - ((p - 1) * rho^2)) / (1 + ((p
- 2) * rho))
+   sigma = sqrt(sigma_squared)
+
+   X <- array(0, dim = c(n, p)) # init arrays
+   X[1, ] <- rnorm(p) # init chains
+
+   for (t in 2:n) {
```

```
+     x = X[t - 1, ]
+     for (i in 1:p) {
+       x_mean_minus = (sum(x) - x[i]) / (p - 1)
+       x[i] = rnorm(1, mean = conditional_mean_coeff * x_mean_minus, sd
= sigma)
+     }
+     X[t, ] = x
+   }
+
+   X
+ }
> X = gibbs_sample(n, p, rho)
> hist(X[, p], breaks = 50, probability = TRUE,
+      main = paste("Marginals of X with p =", p, ", n =", n))
> curve(dnorm(x, 0, 1), add = TRUE, col = "red")
```



Looking at the graph, it follows $N(0, 1)$ pretty well.

(b)

```
> # 1. (b)
>
> library(mnormt)
>
> norm_sample = function(n, p, rho) {
+   # sigma as given in Exercise 7.5
+   I = diag(p)
+   J = matrix(1, p, p)
+   sigma = ((1 - rho) * I + rho * J)
```

```
+
+   rmnorm(n, mean = rep(0, p), varcov = sigma)
+ }
>
> system.time(gibbs_sample(n, p, rho))
   user  system elapsed
  0.003   0.000   0.002
> system.time(norm_sample(n, p, rho))
   user  system elapsed
  0.000   0.000   0.001
```

The rmnorm function seems to be much faster than the Gibbs sampling algorithm, althought this might not be as big of a difference at low iteration count.

2.

We aim to prove that

$$f(x,y) = \frac{f(y|x)}{\int \frac{f(y|x)}{f(x|y)} dy}$$

Let's start from the right-hand side and start evaluating from there:

$$
\begin{aligned}
&= \frac{f(y|x)}{\int \frac{f(y|x)}{f(x|y)} dy} && \text{Given} \\
&= \frac{f(y|x)}{\int \frac{f(y|x)}{\frac{f(x,y)}{f(y)}} dy} && \text{Bayes' theorem} \\
&= \frac{f(y|x)}{\int \frac{f(y|x) \cdot f(y)}{f(x,y)} dy} && \text{Arithmetic} \\
&= \frac{f(y|x)}{\int \frac{f(y|x) \cdot f(y)}{f(y|x) \cdot f(x)} dy} && \text{Definition of conditional probability} \\
&= \frac{f(y|x)}{\int \frac{f(y)}{f(x)} dy} && \text{Arithmetic} \\
&= \frac{f(y|x)}{\frac{1}{f(x)}} && \text{Integration} \\
&= f(y|x) \cdot f(x) && \text{Arithmetic} \\
&= f(x, y) && \text{Definition of conditional probability}
\end{aligned}
$$

Exactly as we sought to prove.